1 Correlations from Mismatched Histories

	A	B	C	D	IDX
A	242	242	221	93	242
B	242	242	221	93	242
C	221	221	228	83	228
D	93	93	83	96	96
IDX	242	242	228	96	250

Table 1: Joint Observations

All five trade 81 out of the 250 days.

a. Building a correlation matrix using pairwise could potentially return a matrix that is not PSD. This is because each entry of a pairwise matrix is estimated from a different sample, so the entries are not required to be mutually consistent.

Building a correlation matrix from complete rows only would be guaranteed PSD. In this case, every entry comes from the same observations.

b. IDX is built as:

$$IDX = 0.4A + 0.3B + 0.2C + 0.1D + e$$

where e is the small tracking error. With the small tracking error, the variance of IDX is $Var(e)$ which is small but above 0. If IDX were exactly the weighted basket, the smallest eigenvalue would be zero. However, with the small tracking error, we expect its smallest eigenvalue to be only slightly above 0. In Cholesky terms, the last diagonal of the root, $\sqrt{A_{55} - \Sigma L_{5k}^2}$, is the variance of IDX not explained by A-D. This is just the tracking error, which is near zero.

This makes the data set highly exposed to the failure in part a). The Week 3 notes explain that pairwise estimation can produce a non-PSD matrix because each correlation is estimated from a different set of observations, so the resulting correlations may be mutually inconsistent. A small inconsistency can push an eigenvalue that is already close to zero below zero. Therefore, I would expect the pairwise correlation matrix for this dataset to be especially vulnerable to having a negative smallest eigenvalue.

c. I would expect the correlations involving D to be the most unreliable, with C-D being the worst overall.

This is because there are few observations. The sample error of a correlation estimate is $(1 - \rho^2)/\sqrt{n}$, so pairs with the smallest joint observations are the noisiest. C-D have the smallest overlap in the table. D also has different time windows compared to the other stocks. For example, A and B share a calendar but D is only listed partway. If the market changed over the year, these entries describe different periods and are not comparable. A newly listed stock may also behave differently in its first months of trading, so the window D does have may not represent its long-run behaviour.

Table 2: Complete Case Correlation Matrix

	A	B	C	D	IDX
A	1.0000	0.3967	0.2466	0.4136	0.8714
B	0.3967	1.0000	0.3169	0.2361	0.6709
C	0.2466	0.3169	1.0000	0.1852	0.5648
D	0.4136	0.2361	0.1852	1.0000	0.5719
IDX	0.8714	0.6709	0.5648	0.5719	1.0000

d. Eigenvalues (ascending): [0.001506 0.548345 0.69332 0.870441 2.886388]

Table 3: Pairwise Correlation Matrix

	A	B	C	D	IDX
A	1.0000	0.4686	0.4309	0.4384	0.8696
B	0.4686	1.0000	0.4630	0.2767	0.7329
C	0.4309	0.4630	1.0000	0.1846	0.7136
D	0.4384	0.2767	0.1846	1.0000	0.6057
IDX	0.8696	0.7329	0.7136	0.6057	1.0000

Eigenvalues (ascending): [-0.011072 0.471422 0.528367 0.857381 3.153902]

Cholesky Attempts: Complete case: succeeded; $max|LL' - A| = 1.11e - 16$
Pairwise: FAILED at IDX: pivot $A_{jj} - sum(L_{jk}^2) = -0.0164275 <= 0$, so its square root is not real

Table 4: Covariance matrix: pairwise correlations

	A	B	C	D	IDX
A	2.7231×10^{-4}	9.2171×10^{-5}	1.3029×10^{-4}	1.6517×10^{-4}	1.7623×10^{-4}
B	9.2171×10^{-5}	1.4206×10^{-4}	1.0112×10^{-4}	7.5286×10^{-5}	1.0728×10^{-4}
C	1.3029×10^{-4}	1.0112×10^{-4}	3.3575×10^{-4}	7.7225×10^{-5}	1.6058×10^{-4}
D	1.6517×10^{-4}	7.5286×10^{-5}	7.7225×10^{-5}	5.2115×10^{-4}	1.6982×10^{-4}
IDX	1.7623×10^{-4}	1.0728×10^{-4}	1.6058×10^{-4}	1.6982×10^{-4}	1.5083×10^{-4}

Table 5: Full-history standard deviations

	Std. Dev.
A	0.0165
B	0.0119
C	0.0183
D	0.0228
IDX	0.0123

e. Calculated portfolio variance, pairwise correlations: -1.7994×10^{-6}
$\implies$ A variance cannot be negative, so this Sigma is not a valid covariance matrix.
Complete-case correlations: 1.0975×10^{-6}

f. See Table 6.

Table 6: Repairs of the pairwise correlation matrix

Repair	Min eigenvalue	Frobenius distance from pairwise	Tracking variance
None (pairwise)	-1.107185×10^{-2}	0	-1.799370×10^{-6}
Rebonato–Jäckel	3.586875×10^{-17}	1.618210×10^{-2}	6.807298×10^{-7}
Higham	$-4.246802 \times 10^{-13}$	1.552000×10^{-2}	6.787649×10^{-7}

g. The covariance matrix from the pairwise correlations is in fact not PSD.

From Week 2, a covariance matrix is PSD if:

$$w'\Sigma w \geq 0 \forall w \in \mathbb{R}^n$$

Note that $Var(w'X) = w'\Sigma w$ which cannot be negative. However, in part e), portfolio variance was calculated to be -1.7994×10^{-6}. Therefore, Σ cannot be PSD.

The eigenvalues say the same thing. The smallest eigenvalue of the pairwise correlation matrix is -0.0111, and the smallest of Σ is -1.93×10^{-6}. A PSD matrix has no negative eigenvalues.

Table 7: Higham changes by entry

Pair	Pairwise	Higham	Change	Days observed
A–IDX	0.8696	0.8625	−0.0070	242
C–IDX	0.7136	0.7088	−0.0048	228
B–IDX	0.7329	0.7287	−0.0042	242
D–IDX	0.6057	0.6020	−0.0037	96
A–C	0.4309	0.4332	+0.0023	221
A–B	0.4686	0.4706	+0.0020	242
A–D	0.4384	0.4402	+0.0018	93
B–C	0.4630	0.4644	+0.0014	221
C–D	0.1846	0.1858	+0.0012	83
B–D	0.2767	0.2777	+0.0011	93

h. The Higham repair moved the entries with IDX the most. All four IDX pairs dropped by 0.004-0.007 while the pairs among A-D rose slightly (0.001-0.002). This is very different than my prediction in c), likely because I factored in number of days behind each entry whereas the repair was not told how many days sit behind each entry. I expected the entries with the fewest observed days (the D pairs) to move the most since they are the least reliable. Higham's method finds the valid correlation matrix (PSD) closest to the pairwise matrix in unweighted Frobenius norm. Every off-diagonal entry is treated equally, whether it rests on 242 days or 83. Where it makes the change depends on the structure of the inconsistency, not on reliability. The pairwise IDX correlations are too high to be consistent with the weak correlations among A-D. Since IDX appears in every conflicting relationship, the cheapest fix is to lower the IDX row and slightly raise the rest.

Table 8: Estimator choice vs. repair choice (Frobenius distances)

Comparison	Frobenius distance
Complete case vs. pairwise	0.4239
Pairwise vs. Higham repair	0.0155
Pairwise vs. Rebonato–Jäckel repair	0.0162
Higham vs. Rebonato–Jäckel	0.0046

i. Complete case vs. pairwise differ by a Frobenius distance of 0.4239, whereas the Higham and Rebonato–Jäckel repairs each move the pairwise matrix by 0.016. The estimator choice therefore moves the matrix roughly 26 times more than the repair choice. The sample holds for the tracking portfolio variance seen in Table 6: it changes by about 38% between the complete-case matrix ($\approx 1.10 \times 10^{-6}$) and the repaired pairwise matrices ($\approx 6.8 \times 10^{-7}$), but by only about 0.3% between the two repairs. The repair is small because the pairwise matrix only mildly violates PSD (smallest eigenvalue 0.0111), so a few entries need to move slightly. The estimators differ more because they use different data: the complete case only uses 81 out of 250 days. This suggests that the choice of estimator matters more for this data as opposed to the repairs.

2 A Volatility Estimate After the Regime Changed

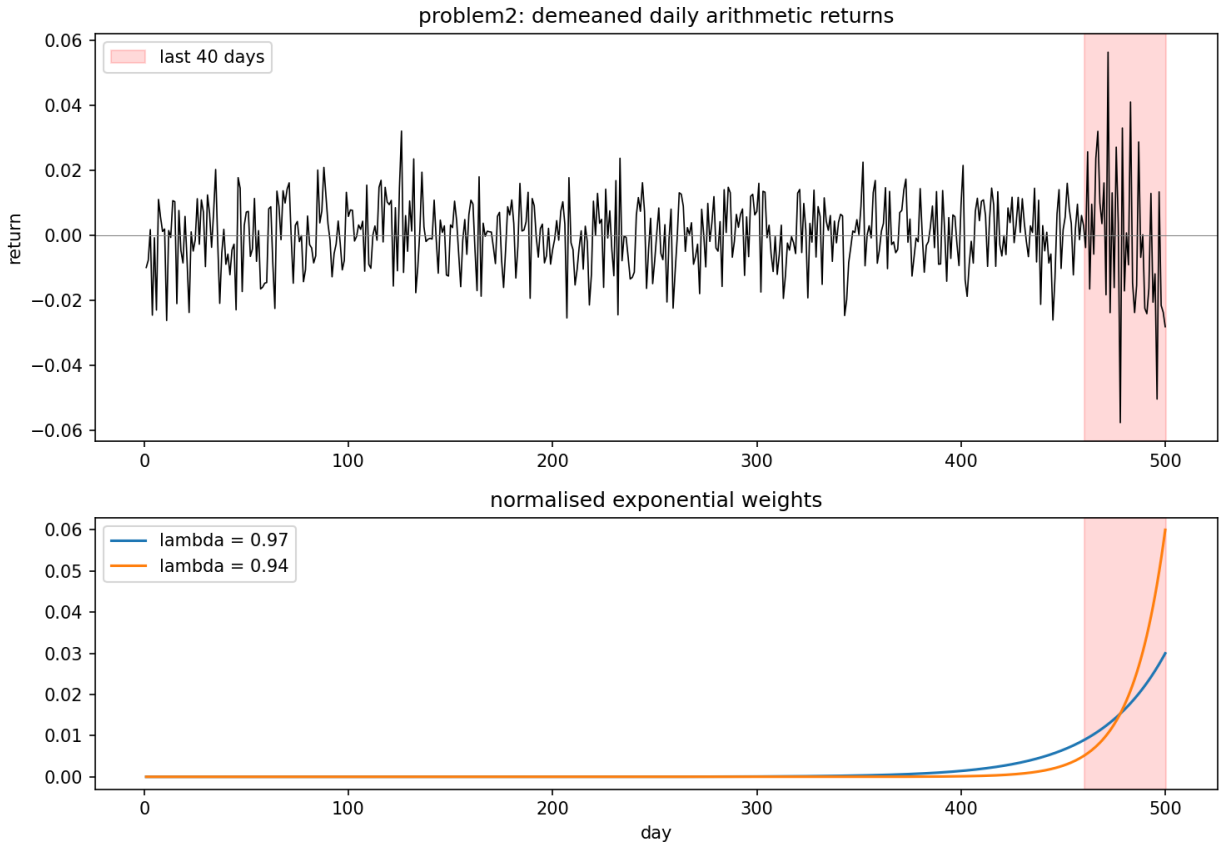


Figure 1: Arithmetic Returns

a. From smallest to large: Equally-weighted normal, historical, Student's t, exponentially weighted $\lambda = 0.97$, exponentially weighted $\lambda = 0.94$.

The equally weighted normal should have a smaller VaR than the historical simulation and Student's t. Model 1 averages out the recent spike in volatility across all 500 trading days and assumes thin normal tails. Historical Simulation (Model 5) and Student's t (Model 4) also average across all 500 days, but they incorporate heavy tails (either empirically or via the fitted degrees of freedom ν), resulting in larger quantile multipliers and a higher VaR than Model 1.

Models 1, 4, and 5 should have a smaller VaR than the exponentially weighted normals. The bottom panel confirms that EWMA places almost all its weight on the most recent observations (the shaded 40-day region). Because the regime change in the last 40 days involves a massive spike in variance, EWMA standard deviations will far exceed the unweighted 500-day standard deviation, yielding significantly higher VaR estimates.

Model 2 should have a smaller VaR than Model 3. The bottom plot explicitly demonstrates that $\lambda = 0.94$ places roughly twice as much weight on the most recent days ($t \approx 500$) compared to $\lambda = 0.97$. Since the highest volatility occurs right at the end of the sample, the smaller decay factor ($\lambda = 0.94$) captures this sharp volatility spike most aggressively, giving it the largest overall VaR estimate.

Table 9: Exponential weighting

λ	$n_{\text{eff}} = \frac{1+\lambda}{1-\lambda}$	n_{eff} (500 obs.)	Half-life (days)	Weight on last 40	Equal-weight share
0.97	65.6667	65.6666	22.7566	0.7043	0.0800
0.94	32.3333	32.3333	11.2023	0.9158	0.0800

b. From the plot in Figure 1, approximately the most recent 40 days look different from the rest. See Table 9 for the effective sample sizes, half life, and fraction of the total weight each λ places on the last 40 days.

c. To predict whether a single regime of normally distributed returns could have produced the first four moments of the full sample, a Jarque-Bera test was used. This way, we can see if skewness and excess kurtosis are statistically indistinguishable from zero. The null hypothesis is that the data is normal.

Table 10: First four moments (full sample)

Statistic	Value
Mean	-2.22045×10^{-19}
Variance	0.000141898
Std. dev.	0.011912
Skewness	-0.156055
Excess kurtosis	2.16594
Tests against normality	
s.e.(skewness) under normal	0.110
s.e.(excess kurtosis) under normal	0.219
Skewness z-score	-1.42
Excess kurtosis z-score	9.89
Jarque–Bera statistic	99.76
Jarque–Bera p-value	2.17×10^{-22}

$JB = \frac{n}{6} \times [S^2 + \frac{(K-3)^2}{4}] = 99.76$
Given the large sample, the Jarque-Bera (JB) statistic follows a Chi square distribution with 2 degrees of freedom. The Chi square critical value at the 1% significance level with 2 degrees of freedom is 9.210. Since JB $>$ 9.210, we reject the null hypothesis that the data is normal.

The high excess kurtosis suggests that a single regime of normally distributed returns could not have produced the first four moments of the sample.

d. See Table 11 for the five VaR estimates in dollars. See Table 12 for the fitted t parameters.

Table 11: One-day 5% VaR on $1,000,000

Method	Vol used[a]	VaR ($)	Rank
1. Normal, equal weight	0.011912	19,594	2
2. Normal, EW $\lambda = 0.97$	0.021047	34,619	4
3. Normal, EW $\lambda = 0.94$	0.023532	38,707	5
4. Student t (MLE)	0.011987	18,950	1
5. Historical simulation	0.011912	20,574	3

Rank: 1 = smallest VaR.

[a] Sample s.d., shown for reference only.

Historical VaR with interpolated quantile: $20,514.

Table 12: Fitted Student t parameters, $X = \mu + \sigma T_\nu$

Parameter	Estimate
μ	2.172388×10^{-4}
σ	0.010076
ν	6.8142
Implied s.d.	0.011987
Implied excess kurtosis	2.1320
Log-likelihood, t	1516.46
Log-likelihood, normal	1506.13

e. My ranking: equally-weighted normal, historical, Student's t, exponentially weighted $\lambda = 0.97$, exponentially weighted $\lambda = 0.94$ Actual ranking: Student's t, equally-weighted normal, historical simulation, exponentially weighted $\lambda = 0.97$, exponentially weighted $\lambda = 0.94$

The result was close to my ranking. What differed was that Student's t actually has the lowest VaR, not equally-weighted normal.

My initial logic relied on the general heuristic that fat-tailed distributions produce larger risk estimates than normal distributions. While that rule of thumb holds true in the extreme tails (e.g., 99% VaR or 99.9% Expected Shortfall), it breaks down at the 95% confidence level due to how probability mass is distributed across a heavy-tailed curve.

Because both Model 1 and Model 4 estimate volatility over the full 500-day sample, their variance baseline is similar. At centre peak, The Student's t distribution is taller and more concentrated around zero. The Student's t distribution has thicker, heavier tails out past 2.5% ($p < 0.025$). To compensate for pushing more probability mass into both the center peak and the extreme outer tails, the Student's t distribution loses mass in the intermediate range between roughly 2.5% and 15%. Because a 95% VaR evaluates the 5% left-tail quantile ($p = 0.05$), the cutoff sits squarely inside this "shoulder" region.

Table 13: Inside each regime

	Old regime (first 460 days)	New regime (last 40 days)	Mixture of normals (92% / 8%)
Std. dev.	0.010178	0.024289	0.011937
Excess kurtosis	−0.202	0.030	2.572
Normal 5% VaR ($)	16,741	39,952	–

New/old s.d. ratio: 2.39×.

Fitted t: $\sigma = 0.010076$ (close to old-regime s.d. 0.010178), $\nu = 6.81$.

f. The standard deviation is 0.0102 in the old regime (first 460 days) and 0.0243 in the new regime (last 40 days), 2.39 times higher (Table 13). Within each regime, returns show essentially no excess kurtosis (−0.20 and 0.03), so each regime looks normal on its own.

A mixture of two zero-mean normals gives a s.d. of 0.0119 and an excess kurtosis of 2.57, close to the full-sample values of 0.0119 and 2.17. Most days come from the calm regime, which makes a tall, narrow center, and the few high-volatility days fill the tails. The data have fat tails because the volatility shifted, not because daily returns are fat-tailed.

The fitted t describes this mixture, not a single fat-tailed distribution. Its scale ($\sigma = 0.0101$) almost matches the old-regime s.d. of 0.0102, so the body of the t is the calm regime. Its degrees of freedom ($\nu = 6.81$) imply an excess kurtosis of $6/(\nu - 4) = 2.13$, almost exactly the sample value of 2.17. The t chose its tails to match the kurtosis created by mixing the two regimes. Its much better fit than the normal ($\text{LR} = 2(1516.46 - 1506.13) = 20.66 > \chi^2_{1,\,0.01} = 6.63$) reflects the regime shift, not fat tails within either regime. In effect, the t treats a temporary change in volatility as a permanent feature of the tails.

Table 14: EW VaR gap vs. estimation noise, s.e.$(\hat{\sigma}) = \hat{\sigma}/\sqrt{2n_{\text{eff}}}$

λ	$\hat{\sigma}$	n_{eff}	s.e.$(\hat{\sigma})$	s.e.(VaR) (\$)
0.97	0.021047	65.67	0.001837	3,021
0.94	0.023532	32.33	0.002926	4,813
Gap: VaR(0.94) − VaR(0.97) = \$4,088				
Gap / s.e.(0.97)			1.35	
Gap / s.e.(0.94)			0.85	
Gap / $\sqrt{\text{s.e.}_{0.97}^2 + \text{s.e.}_{0.94}^2}$			0.72	

g. No, the gap is not larger than the estimation noise. From Table 14, the difference between the two 1-day VaR estimates is \$4,088. The joint estimation noise (assuming independence) is $\sqrt{\text{s.e.}_{0.97}^2 + \text{s.e.}_{0.94}^2} = \$4088 \div 0.72 = \$5,683$.

For a trading limit, I would use $\lambda = 0.94$ because trading limits need to react quickly to abrupt regime shifts and sudden spikes in market volatility. A lower decay factor ($\lambda = 0.94$) has an effective sample size of $n_{\text{eff}} \approx 32$ days, placing much higher weight on recent events. This forces traders to scale back positions faster when risk spikes.

For a capital number, I would use $\lambda = 0.97$ because capital requirements are designed to buffer against loss over longer horizons and should remain stable to prevent rapid, costly capital re-allocations. A higher decay factor ($\lambda = 0.97$) doubles the effective sample size to $n_{\text{eff}} \approx 66$ days, yielding a significantly lower standard error (\$3,021 vs. \$4,813).

3 When Diversification Raises VaR

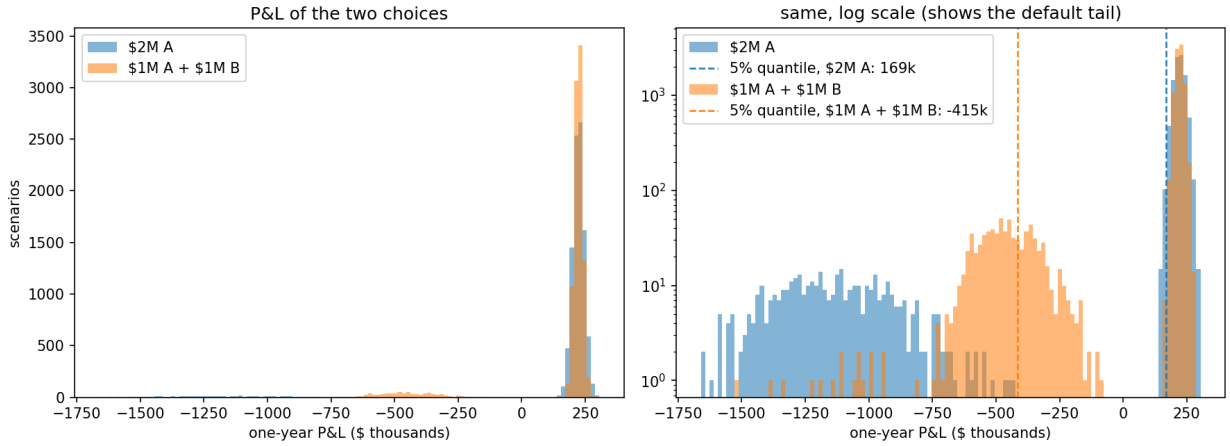


Figure 2: P&L Distribution

Table 15: First four moments of each bond's one-year return

Statistic	Bond A	Bond B
Mean	0.0837092	0.0850113
Variance	0.0184926	0.0174509
Std. dev.	0.135988	0.132102
Skewness	−4.82325	−4.91552
Excess kurtosis	22.0936	23.0865

a. The skewness and excess kurtosis values for both bonds are far from zero. This suggests that neither return follows a normal distribution.

A Normal VaR model assumes zero skewness and zero excess kurtosis. This would lead to severe underestimation of tail risk. A normal distribution spreads risk symmetrically across both left and right tails. In reality, default risk generates a long, fat left tail of catastrophic losses.

It also leads to inaccurate quantile estimates. Normal VaR relies on standard multipliers (e.g., 1.645σ for 95% VaR). Due to the high excess kurtosis ($K > 22$), the true empirical loss tail extends far beyond normal multiplier bounds, causing Normal VaR to significantly underestimate actual Value at Risk and tail risk.

b. The diversified portfolio of A&B should have a larger VaR than the concentrated one. A itself loses more than 20% in 4.06% of scenarios. Since $4.06\% < 5\%$, the 95th percentile loss threshold (95% VaR) falls outside the severe default loss event. Thus, its 95% VaR remains relatively low. A&B together result in at least one of them losing more than twenty percent 7.82% of the time. Because $7.82\% > 5\%$, the 5% tail threshold now penetrates directly into the default event zone. As a result, the diversified portfolio breaches the 5% loss probability cutoff much more frequently, leading to a larger 95% VaR.

On the other hand, the diversified portfolio should have a smaller ES than the concentrated one. In a concentrated portfolio, when Bond A suffers > 20% loss (4.06% of scenarios), the portfolio suffers a full 100% loss exposure on the defaulted asset. Whereas with a diverse portfolios, because joint defaults are extremely rare (0.17%), the two bonds default almost independently. In most severe loss scenarios (7.82% - 0.17% = 7.65%), only one bond defaults while the other pays off. Splitting $2,000,000 across both bonds cuts the severity of single-default losses in half, lowering the average magnitude of losses in the 5% tail and producing a smaller ES.

Table 16: Scenarios with a loss of more than 20%

Event	Scenarios	Share of scenarios
A loses > 20%	406	4.06%
B loses > 20%	393	3.93%
At least one loses > 20%	782	7.82%
Both lose > 20%	17	0.17%

Table 17: Risk at $\alpha = 5\%$, one year, in $ (positive = loss)

Position	Hist. VaR	Hist. ES	Normal VaR	Normal ES	Mean P&L	s.d. P&L
$1M A	−84,531	443,628	139,971	196,794	83,709	135,988
$1M B	−86,011	419,587	132,277	187,477	85,011	132,102
$2M A	−169,061	887,257	279,941	393,589	167,418	271,975
$1M A + $1M B	414,767	541,997	143,466	222,774	168,721	189,796

c. See Table 17

d. See Table 17

e. See Table 18.

Table 18: Risk at $\alpha = 1\%$, one year, in \$ (positive = loss)

Position	Hist. VaR	Hist. ES	Normal VaR	Normal ES	Mean P&L	s.d. P&L
\$1M A	650,545	709,548	232,645	278,727	83,709	135,988
\$1M B	641,248	700,805	222,303	267,068	85,011	132,102
\$2M A	1,301,090	1,419,096	465,291	557,454	167,418	271,975
\$1M A + \$1M B	598,237	716,408	272,811	337,127	168,721	189,796

Table 19: Subadditivity: is $q(\$1\text{M A} + \$1\text{M B}) \leq q(\$1\text{M A}) + q(\$1\text{M B})$?

Measure	$q(X)$	$q(Y)$	$q(X)+q(Y)$	$q(X+Y)$	Benefit	Subadditive
Hist. VaR 5%	−84,531	−86,011	−170,542	414,767	−585,309	No
Hist. ES 5%	443,628	419,587	863,216	541,997	321,218	Yes
Normal VaR 5%	139,971	132,277	272,247	143,466	128,781	Yes
Hist. VaR 1%	650,545	641,248	1,291,793	598,237	693,556	Yes
Hist. ES 1%	709,548	700,805	1,410,353	716,408	693,945	Yes

X = \$1M A, Y = \$1M B. Benefit $= q(X) + q(Y) - q(X+Y)$; negative means combining added risk.

Table 20: Concentrated (\$2M A) vs. diversified (\$1M A + \$1M B): which does each measure prefer?

Measure	Concentrated	Diversified	Prefers
Hist. VaR 5%	−169,061	414,767	Concentrated
Hist. ES 5%	887,257	541,997	Diversified
Normal VaR 5%	279,941	143,466	Diversified
Hist. VaR 1%	1,301,090	598,237	Diversified
Hist. ES 1%	1,419,096	716,408	Diversified
Normal VaR 1%	465,291	272,811	Diversified
s.d. of P&L	271,975	189,796	Diversified

Lower risk is preferred. Mean P&L of the two choices differs by only \$1,302.

f. Subadditivity holds for historical ES and normal VaR but not for historical VaR.

For individual positions, each bond defaults roughly 4% of the time. Because $4\% < 5\%$, the 95th percentile cutoff ($p = 0.05$) for an individual \$1M bond position falls in the non-default region, resulting in a negative loss estimate ($-\$84,531$ and $-\$86,011$, meaning a gain). For the combined position, holding both bonds creates two independent chances to default. The probability of at least one bond defaulting rises to 7.82%. Because $7.82\% > 5\%$, the 5% tail cutoff now crosses directly into the single-default loss zone, raising the portfolio VaR to $\$414,767$. Combining two assets spreads the total default frequency across the 5% threshold, causing VaR to report that diversifying increases risk.

When a single bond defaults, an individual \$1M position suffers a full concentrated default loss. In the combined portfolio, joint defaults are extremely rare (0.17%). In nearly all tail scenarios (7.65%), only one bond defaults while the other pays off. Because ES averages the entire tail, halving the exposure to a single default reduces the severity of the worst 5% of scenarios, yielding a subadditive risk reduction of $\$321,218$.

g. Normal VaR favours the diversified portfolio (\$1,000,000 in A + \$1,000,000 in B). Benefit of switching from concentrated to diversified is $\$279,941 - \$143,466 = \$136,475$ reduction in VaR. Because Normal VaR drops significantly for the combined portfolio, a decision-maker relying solely on Normal VaR would conclude that diversification strongly reduces downside risk.

No, normal VaR arrives at the correct qualitative conclusion (that diversification helps), but for completely wrong and misleading reasons. Normal VaR relies strictly on standard deviation (σ) and

assumes a bell-shaped, symmetric Gaussian distribution. Because combining the two bonds reduces portfolio standard deviation from $\$271,975$ down to $\$189,796$ (due to imperfect correlation), Normal VaR blindly multiplies this lower portfolio standard deviation by the 1.645 Gaussian factor ($1.645 \times \$189,796 \approx \$143,466$). Both bonds have severe negative skewness (≈ -4.8 to -4.9) and massive excess kurtosis (≈ 22 to 23) due to default risk. Normal VaR completely ignores these higher moments.s

h. At 1%, subaddivity now holds across both VaR and ES.

At the 5% confidence level, the historical VaR of the combined portfolio violates subadditivity because its $\$414,767$ risk measure exceeds the $-\$170,542$ sum of the individual VaRs. This occurs because individual bond defaults happen 4.06% and 3.93% of the time. Since both individual default rates fall below 5%, each single-bond 5% cutoff remains in the non-default zone. However, combining the bonds raises the chance of at least one default to 7.82%, crossing the 5% threshold and causing the combined VaR to spike into the single-default loss range.

At the 1% confidence level, subadditivity is restored because the 1% cutoff sits deep inside the default region for both individual positions and the combined portfolio. An individual \$1M position at 1% reflects a full default loss, whereas the combined portfolio's 1% quantile reflects only a single default because joint defaults are extremely rare at 0.17%. As a result, the portfolio's 1% VaR of $\$598,237$ is far less than the individual sum of $\$1,291,793$, generating a $\$693,556$ diversification benefit.

This shift demonstrates that VaR's failure to recognize diversification depends heavily on the chosen confidence level relative to an asset's default probability. In heavy-tailed, highly skewed portfolios, VaR creates a "cliff effect" by evaluating a single percentile cutoff rather than the full loss tail. Moving to Expected Shortfall (ES) resolves this vulnerability entirely, maintaining subadditivity across both 5% and 1% levels by averaging all losses beyond the threshold.

4 Gaussian or t Copula

Table 21: First four moments of each series

Statistic	X1	X2	X3
Mean	0.000256585	−0.000252381	0.0008898
Variance	0.00014147	0.00026142	0.000100377
Std. dev.	0.0118941	0.0161685	0.0100188
Skewness	−0.0342258	−0.61504	0.0740662
Excess kurtosis	−0.117133	7.12496	2.87409

a. I would fit X1 into a normal distribution because its excess kurtosis is close to zero. Since X2 and X3 have large excess kurtosis, I would fit them to a t distribution.

Table 22: Effect of dropping the single most extreme day (day 951) of each series

	X1		X2		X3	
	All	Drop	All	Drop	All	Drop
Skewness	−0.0342	−0.0055	−0.6150	0.0559	0.0741	0.3132
Excess kurtosis	−0.1171	−0.1876	7.1250	1.7247	2.8741	1.7307
Day 951 return	−0.04000		−0.14291		−0.06124	

Day 951 is row 953 of the CSV. "Drop" excludes that day.

The kurtosis of X2 looks out of place. It is extremely high at 7.12. Dropping the observation for Day 951, we can see in Table 22 that this drastically reduces the excess kurtosis for the returns of X2, from 7.12 to 1.72. Skew also changed directions, from -0.62 to 0.06. For X1 and X3, skew and excess kurtosis barely changed.

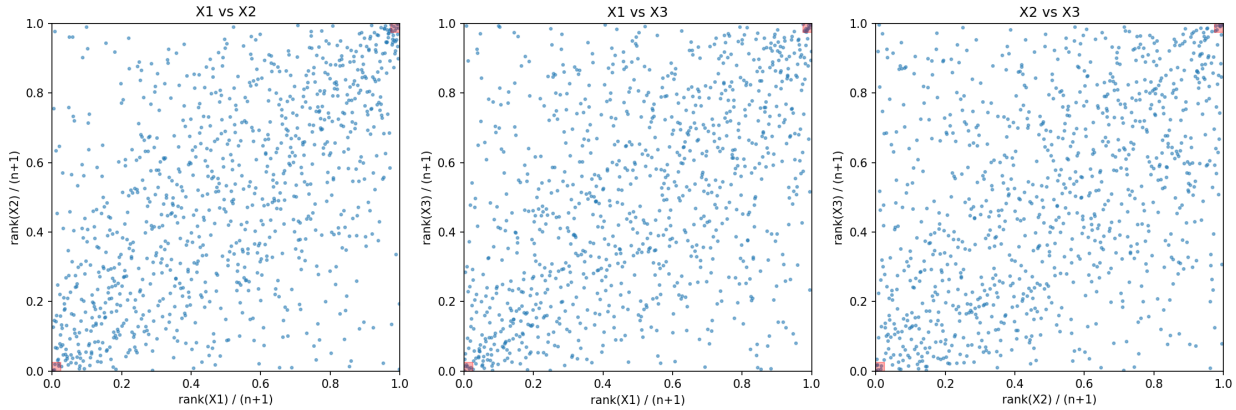


Figure 3: Plot of Assets' Ranks

Table 23: Days on which both series are in their own worst or best 2.5%

Pair	Lower tail (worst 2.5%)	Upper tail (best 2.5%)
X1–X2	6	8
X1–X3	7	12
X2–X3	6	9

b.

$$\text{Expected Days} = N \times P(U_1 \leq 0.025) \times P(U_2 \leq 0.025) = 1000 \times 0.025 \times 0.025 = 0.625 \text{ days}$$

The plots and counts do suggest that the joint tails are more connected than the middle of the distribution. The empirical joint tail counts (6–12 days) are roughly 10 to 19 times higher than the independence benchmark of 0.625 days, showing strong joint tail clustering. Looking at Figure 3, points cluster heavily in the corners $(0, 0)$ and $(1, 1)$ compared to the center. Consequently, I predict the t copula will win. The t copula features an extra degrees-of-freedom parameter (ν) that explicitly allows for positive joint tail dependence ($\lambda_L = \lambda_U > 0$). It naturally concentrates extra probability density in the corners to model simultaneous joint crashes and joint booms.

c. See Table 24.

Table 24: Margin Fits

Series	AICc normal	AICc t	AICc(n) − AICc(t)	t μ	t σ	t ν	Normal s.d.	Chosen
X1	−6022.54	−6017.38	−5.156	2.686×10^{-4}	0.011538	30.00	0.011888	Normal
X2	−5408.49	−5517.88	109.385	-1.736×10^{-4}	0.012392	4.847	0.016160	t
X3	−6365.69	−6428.02	62.324	7.457×10^{-4}	0.008182	6.103	0.010014	t

d. See Tables 25 and 26.

Table 25: Copula correlation matrix from Kendall's τ, $R_{ij} = \sin\left(\frac{\pi}{2}\tau_{ij}\right)$

	X1	X2	X3
X1	1.0000	0.5833	0.5029
X2	0.5833	1.0000	0.4644
X3	0.5029	0.4644	1.0000

Table 26: Copula fits

Copula	Log-likelihood	$\hat{\nu}$	k	AICc	BIC
Gaussian	350.6099	∞	3	−695.1956	−680.4965
t	452.4186	3.6051	4	−896.7969	−877.2061

AICc and BIC both favor the t copula. A $\hat{\nu}$ of about 3.6 implies strong tail dependence, which fits the joint-tail counts from part (b).

e. See Table 27.

Table 27: Portfolio VaR and ES, one day, in $ (positive = loss); 100,000 simulated days per copula

Method	VaR 5%	ES 5%	VaR 1%	ES 1%
Gaussian copula	49,738	66,956	76,889	94,842
t copula	49,240	68,749	79,583	101,888
Historical	47,939	65,699	70,092	102,424

f. See Table 28.

Table 28: Implied joint-tail days per 1,000 days

	Gaussian copula		t copula	
Pair	Lower	Upper	Lower	Upper
X1–X2	5.92	6.37	9.28	8.91
X1–X3	4.71	4.66	8.01	8.26
X2–X3	4.32	4.28	7.74	7.47

Table 29: Which copula wins?

Criterion	Value
Log-likelihood: $t-$ Gaussian	101.809
AICc: Gaussian $- t$	201.601
BIC: Gaussian $- t$	196.710

Positive differences favor the t copula.

g. As shown in Table 29, the Student's t copula wins decisively across all model selection criteria. The t copula increases log-likelihood by +101.809 units over the Gaussian copula ($\ell_t - \ell_{\text{Gauss}} = 101.809$). The AICc difference favoring the t copula ($\text{AICc}_{\text{Gauss}} - \text{AICc}_t$) is 201.601. The BIC difference favoring the t copula ($\text{BIC}_{\text{Gauss}} - \text{BIC}_t$) is 196.710.

Table 30: Joint-tail days: data vs. implied by each copula

	Data		Gaussian		t	
Pair	Lower	Upper	Lower	Upper	Lower	Upper
X1–X2	6	8	5.92	6.37	9.28	8.91
X1–X3	7	12	4.71	4.66	8.01	8.26
X2–X3	6	9	4.32	4.28	7.74	7.47
Total	19	29	14.95	15.31	25.03	24.64

The Gaussian copula severely underestimates joint tail events. It predicted only ~ 30 total tail days versus the 48 observed in reality because it lacks asymptotic tail dependence. The t copula captures

total joint extreme co-movements much better ($\sim$ 49.7 predicted vs. 48 actual). While it slightly overpredicts the lower tail (25.03 vs. 19) and slightly underpredicts the upper tail (24.64 vs. 29), its symmetric tail dependence parameter (ν) provides a far superior fit to the joint extreme behavior compared to the Gaussian copula.

Table 31: Moving from the Gaussian to the t copula, one day, in \$ (positive = loss)

Measure	Gaussian	t	$t-$ Gaussian (\$)	Relative change
VaR 5%	49,738	49,240	−498	−1.00%
ES 5%	66,956	68,749	+1,793	+2.68%
VaR 1%	76,889	79,583	+2,694	+3.50%
ES 1%	94,842	101,888	+7,045	+7.43%

Largest move: ES 1%.

h. Switching from the Gaussian to the t copula moved 1% ES the most. On the other hand, the 5% VaR moved the least. This is because of where the copulas allocate their probability mass. 1% ES averages all losses beyond the 1% threshold, directly capturing the severe joint default and drawdown events modeled by the t copula.

The primary mathematical difference between a Student's t copula and a Gaussian copula is joint asymptotic tail dependence. A t copula increases the probability of extreme, simultaneous joint market crashes (the far outer tail). The 5% VaR evaluates a single percentile cutoff sitting at $p = 0.05$. Because $p = 0.05$ sits in the intermediate "shoulder" of the joint distribution, the stronger joint tail clustering of the t copula has not yet taken full effect at that cutoff. In fact, to accommodate the heavier extreme tails ($p \leq 1\%$), probability mass is pulled away from the intermediate shoulders, causing 5% VaR to remain virtually unchanged (or drop slightly). VaR only checks a single threshold point. Since the t copula's joint crash behavior happens beyond $p = 0.05$, 5% VaR completely misses the heavy-tail co-movements.

Table 32: Daily contributions to $\ell_t - \ell_{\text{Gauss}}$

Days	Count	Contribution	Share of total	Mean per day
No series in its outer 5%	795	50.407	49.5%	+0.0634
At least one in its outer 5%	205	51.401	50.5%	+0.2507
Total	1,000	101.809	100%	+0.1018

The t copula fits better on 61.0% of middle days.

i. 49.5% of the total comes from days where no series sits in its outer 5% on either side.

This tells me that the information criteria score overall fit across the entire distribution, not just tail risk. AICc and BIC evaluate the global joint density across all 1,000 observations. They are not purely "tail risk" metrics.

Table 33: Tail dependence for the most correlated pair, X1–X2 ($\rho = 0.5833$)

Copula	ν	λ (each tail)
t	3.605	0.3251
Gaussian	∞	0.0000

$\lambda_t = 2\,t_{\nu+1}\left(-\sqrt{(\nu+1)\frac{1-\rho}{1+\rho}}\right)$; the t copula is symmetric, so $\lambda_L = \lambda_U$.

j. The fundamental structural flaw of the Gaussian copula is its assumption that $\lambda_L = 0$ for every $\rho < 1$. The t copula correctly captures the 32.51% chance of simultaneous extreme tail co-movements seen in financial markets.

5 Model Based Simulation and Residual Correlation

a. Let $P = w_A r_A + w_B r_B$. By Week 2's fact 7,

$$\mathrm{Var}(P) = \sum_i \sum_j w_i w_j \,\mathrm{cov}(r_i, r_j) = w_A^2 \,\mathrm{Var}(r_A) + w_B^2 \,\mathrm{Var}(r_B) + 2\, w_A w_B \,\mathrm{cov}(r_A, r_B).$$

With $r_i = \alpha_i + \beta_i r_M + \epsilon_i$, the alphas drop out (fact 5), and expanding by fact 6,

$$\mathrm{cov}(r_A, r_B) = \beta_A \beta_B \sigma_M^2 + \beta_A \,\mathrm{cov}(r_M, \epsilon_B) + \beta_B \,\mathrm{cov}(r_M, \epsilon_A) + \mathrm{cov}(\epsilon_A, \epsilon_B) = \beta_A \beta_B \sigma_M^2 + \mathrm{cov}(\epsilon_A, \epsilon_B),$$

since the regression errors are uncorrelated with the market. Each stock's own variance, $\mathrm{Var}(r_i) = \beta_i^2 \sigma_M^2 + \mathrm{Var}(\epsilon_i)$, does not depend on the shortcut, so the shortcut only changes the cross term

$$2\, w_A w_B \,\mathrm{cov}(r_A, r_B) = 2\, w_A w_B \beta_A \beta_B \sigma_M^2 + 2\, w_A w_B \,\mathrm{cov}(\epsilon_A, \epsilon_B).$$

Setting $\mathrm{cov}(\epsilon_A, \epsilon_B) = 0$ removes the last term. A shared industry makes $\mathrm{cov}(\epsilon_A, \epsilon_B) > 0$, so:

- `P1` (long A, long B): $w_A w_B > 0$, so the shortcut drops a *positive* term and **understates** the VaR.
- `P2` (long A, short B): $w_A w_B < 0$, so the shortcut drops a *negative* term and **overstates** the VaR.

b. See Table 34.

Table 34: OLS of each stock on the market

Stock	α	β	Resid. s.d.	R^2
A	2.050×10^{-4}	1.0353	0.012901	0.4181
B	3.148×10^{-4}	0.9354	0.011550	0.4226

$\mathrm{corr}(e_A, e_B) = 0.6983$; $\mathrm{corr}(A, B)$ raw returns $= 0.8251$; market s.d. $= 0.010553$.
Resid. s.d. is the regression standard error ($n - 2$ degrees of freedom).

c. See Tables 35 and 36.

Table 35: Simulated 5% VaR, 100,000 days, in $ (positive = loss)

Portfolio	Full residual cov.	Independent residuals
P1 (long A, long B)	50,307	44,435
P2 (long A, short B)	15,881	28,616

Table 36: Residual covariance used for the full run

	A	B
A	1.6611×10^{-4}	1.0385×10^{-4}
B	1.0385×10^{-4}	1.3314×10^{-4}

d. See Tables 37 and 38.

Table 37: Sample covariance of A and B

	A	B
A	2.8546×10^{-4}	2.1169×10^{-4}
B	2.1169×10^{-4}	2.3058×10^{-4}

Table 38: Delta-normal 5% VaR from the sample covariance, in $ (positive = loss)

Portfolio	VaR 5%
P1 (long A, long B)	50,415
P2 (long A, short B)	15,834

e. Yes, the shortcut moved each portfolio in the way I predicted. For P1, the shortcut lowered the VaR from $\$50,307$ to $\$44,435$ (understating VaR by $\$5,872$). For P2, the shortcut raised the VaR from $\$15,881$ to $\$28,616$ (overstating VaR by $\$12,735$).

The magnitudes of error:

- P1 (Long/Long): The relative error is $\frac{44{,}435-50{,}307}{50{,}307} \approx -11.7\%$.
- P2 (Long/Short): The relative error is $\frac{28{,}616-15{,}881}{15{,}881} \approx +80.2\%$.

So P2 (Long/Short) is far more exposed to the independence assumption.

In a long/long portfolio (P1), the primary driver of risk is systemic market risk ($\beta_A\beta_B\sigma_M^2$), which remains active under both models. The residual covariance term is additive to a large market variance baseline, so dropping it causes only a modest percentage change.

In a long/short portfolio (P2), market exposure is largely hedged out because the long and short market positions offset each other ($\beta_A - \beta_B \approx 1.0353 - 0.9354 = 0.0999 \approx 0$). Because systemic market variance is reduced to near zero, the residual terms dominate the portfolio variance. Ignoring positive residual covariance strips away the main source of remaining risk mitigation (the hedging benefit between residual errors), nearly doubling the reported VaR.

f. The two approaches yield nearly identical results (differing only by negligible simulation noise across 100,000 runs).

Under OLS regression with linear relationships and joint normal errors, the total sample covariance matrix of returns ($\mathbf{\Sigma}_{\text{sample}}$ in Table 37) decomposes exactly into the market component and the residual covariance matrix. Since OLS residuals are orthogonal to market returns by construction, a linear combination of joint normal variables (the factor model simulation) is mathematically identical to a linear combination of returns using the sample covariance matrix. Therefore, Delta-Normal VaR and full-covariance simulation match perfectly.

The two stop agreeing as soon as the simulation is no longer a draw from that same joint normal:

- Any piece is not normal. If the errors (or the market) are drawn from a fitted t, the simulated P&L is no longer normal and its 5% quantile is no longer $1.645\sigma_p$. Joining non-normal pieces requires a copula.
- The residual covariance is not the full sample one. Setting the off diagonal to zero (part c) breaks the equality: `P1` drops to $\$44,435$ and `P2` rises to $\$28,616$. The same happens if Σ_ϵ or σ_M^2 is estimated on a different window or with exponential weights.
- The betas are not OLS on this sample. With imposed betas, or betas from another period, the errors are correlated with the market in this sample, so the cross terms in the covariance do not vanish and $\beta\beta^\top\sigma_M^2 + \Sigma_\epsilon$ no longer reproduces Σ_{sample}.